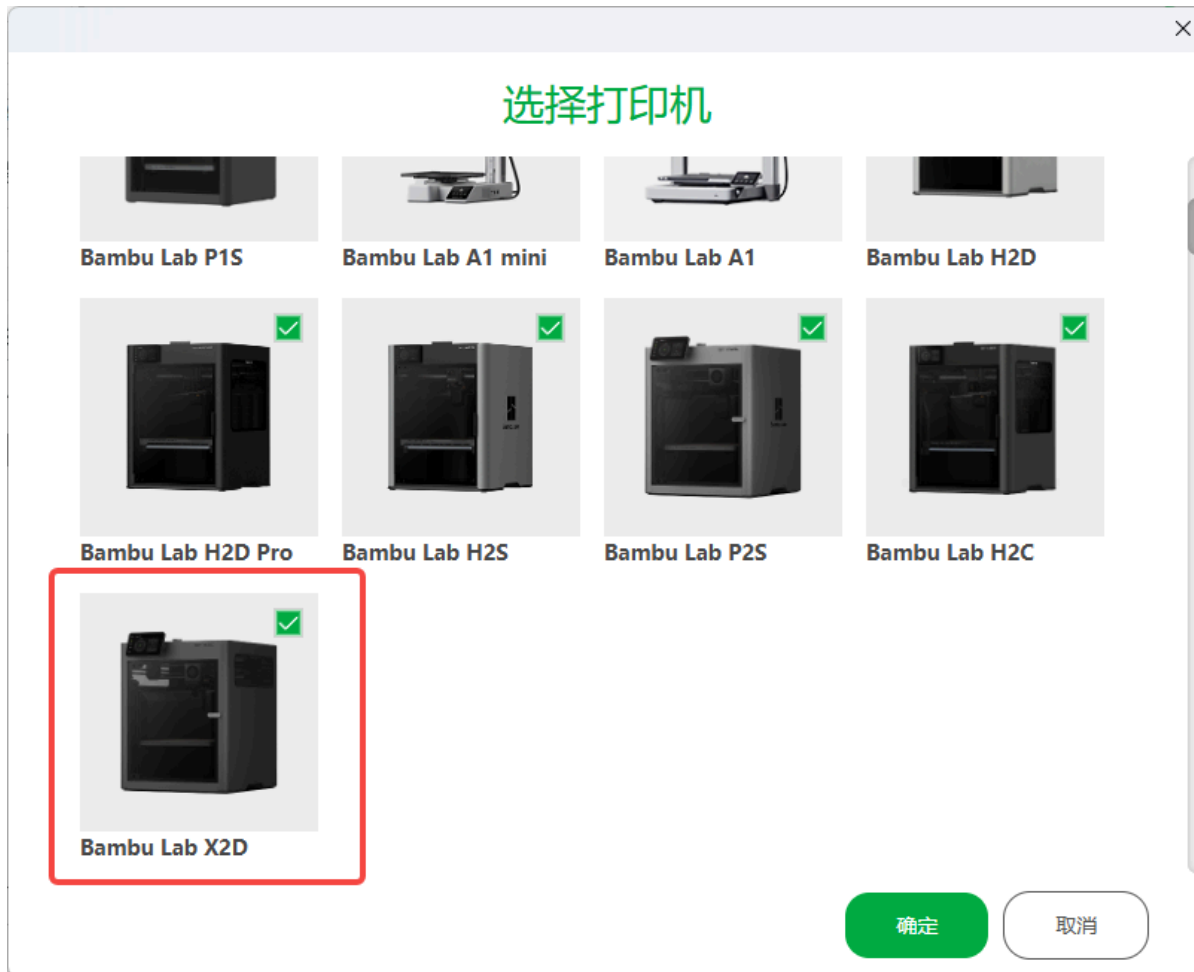


Bambu Studio V2.5.3 Release Note

V2.5.3 adds support for the Bambu Lab X2D, along with new features, improvements, and bug fixes.



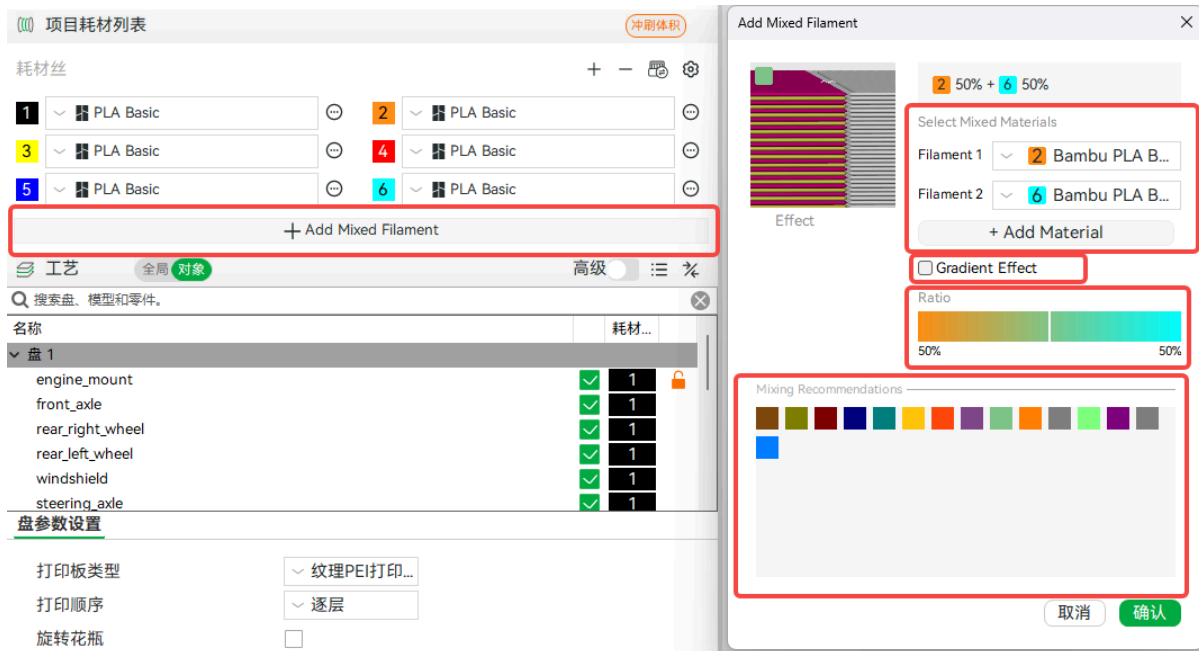
Features

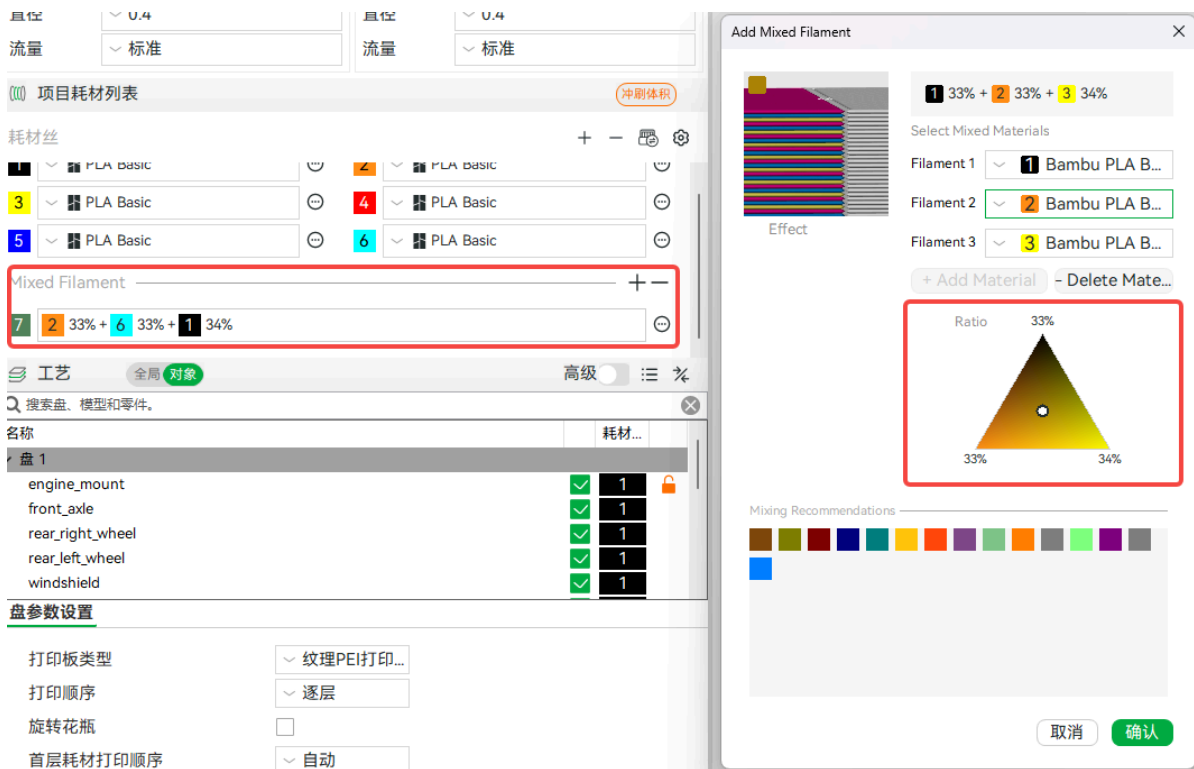
Color Mixing

The Color Mixing feature allows you to mix multiple filaments in the same print to create new colors by adjusting the ratio. To use it, click "Add Mixed Filament" at the bottom of the filament list to configure them in the pop-up dialog.



- ▶ Supports mixing 2 or 3 filaments of the same type.
- ▶ Mixing modes:
 - ▶ Normal mode: Adjust the ratio of two or three colors using the color bar or color triangle in the ratio panel, or select directly from the recommended colors below. With the Bambu Lab CMYK filament kit, a wide range of colors can be achieved – for example, all the boats shown in the left image were printed using the CMYK filament kit on a single plate.
 - ▶ Gradient mode: Create gradient color transitions between two colors for rich transitional effects, as shown in the right image.





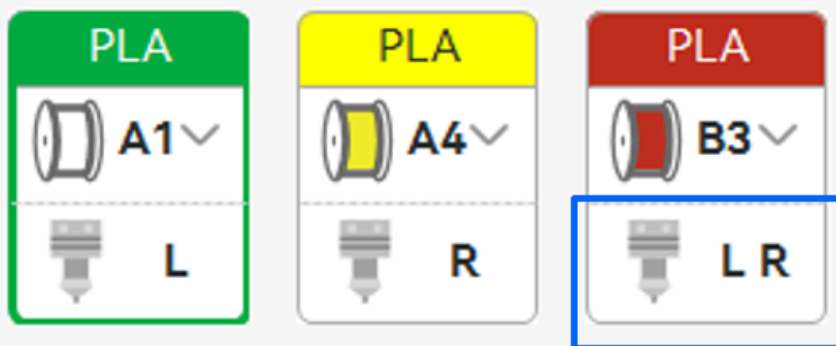
Note:

1. This feature is only suitable for near-vertical walls. It is not recommended for sloped surfaces or top/bottom surface color layering.
2. Single-nozzle printers are not recommended for this feature due to frequent color changes and increased filament consumption. A multi-nozzle printer is recommended for better results.
3. With a 0.4mm nozzle, a base layer height of 0.12mm and a mixed layer height of 0.2mm are recommended for more uniform color mixing results. Avoid extreme layer height ratios, as excessively small layer heights may trigger melt fracture and degrade print quality.
4. The color preview model for layered colors is not yet fully accurate. Users with high color requirements are advised to print a small test piece first and fine-tune the settings based on the observed results.

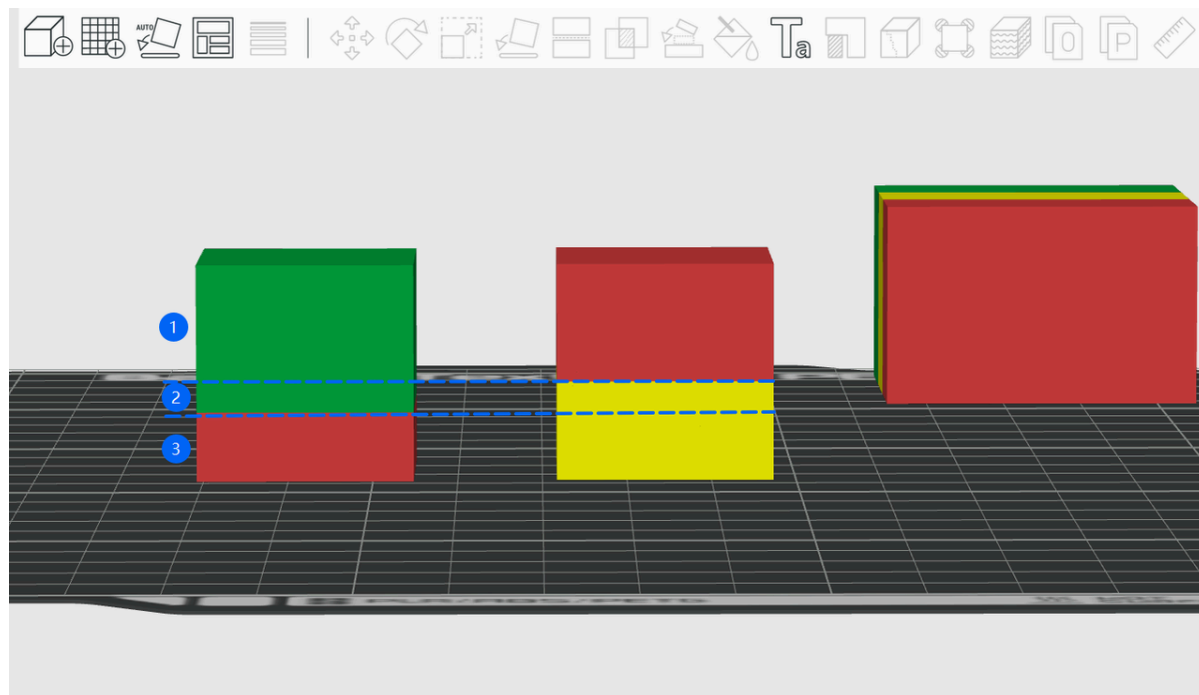
Filament Track Switch Support

The Filament Track Switch enables filament from the AMS to be fed into both the left and right nozzles of multi-extruder printers (H2C/X2D).

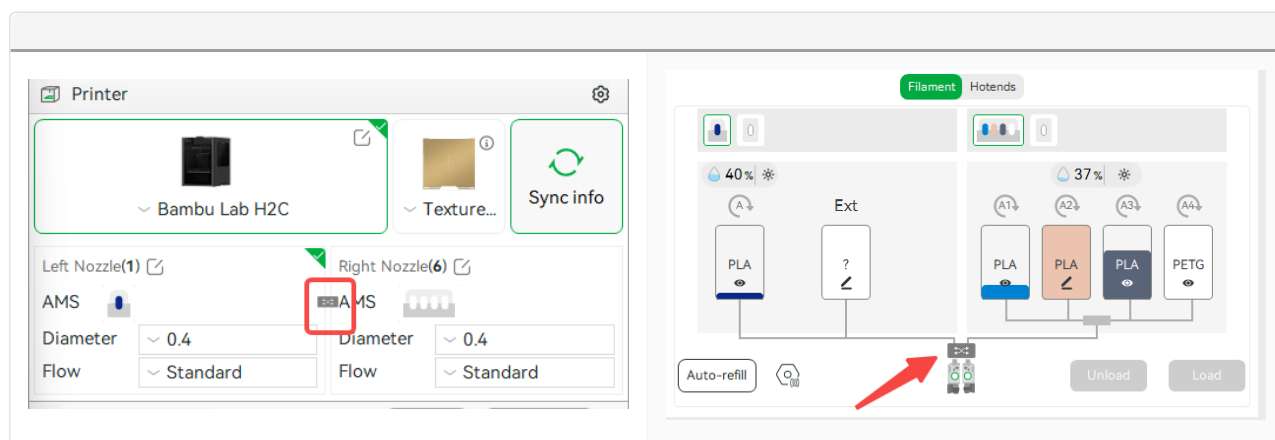
Filament



Slicing allows a single filament to be printed using different nozzles in this mode, which effectively reduces unnecessary flushing and significantly improves multi-color printing efficiency.



UI:



Tip: The Filament Track Switch will only appear in printer presets and the Devices page after syncing a device with the Filament Track Switch installed via the Sync button. Smart assignment mode will also be enabled in slicing mode.

☒ Filament-Saving Mode

Generates filament grouping for the left nozzle and right nozzle based on the most filament-saving principles to minimize waste

☐ Custom Mode

Manually assign filament to the left nozzle or right nozzle

Enable smart filament assign: Assign one filament to multiple nozzles to maximize savings



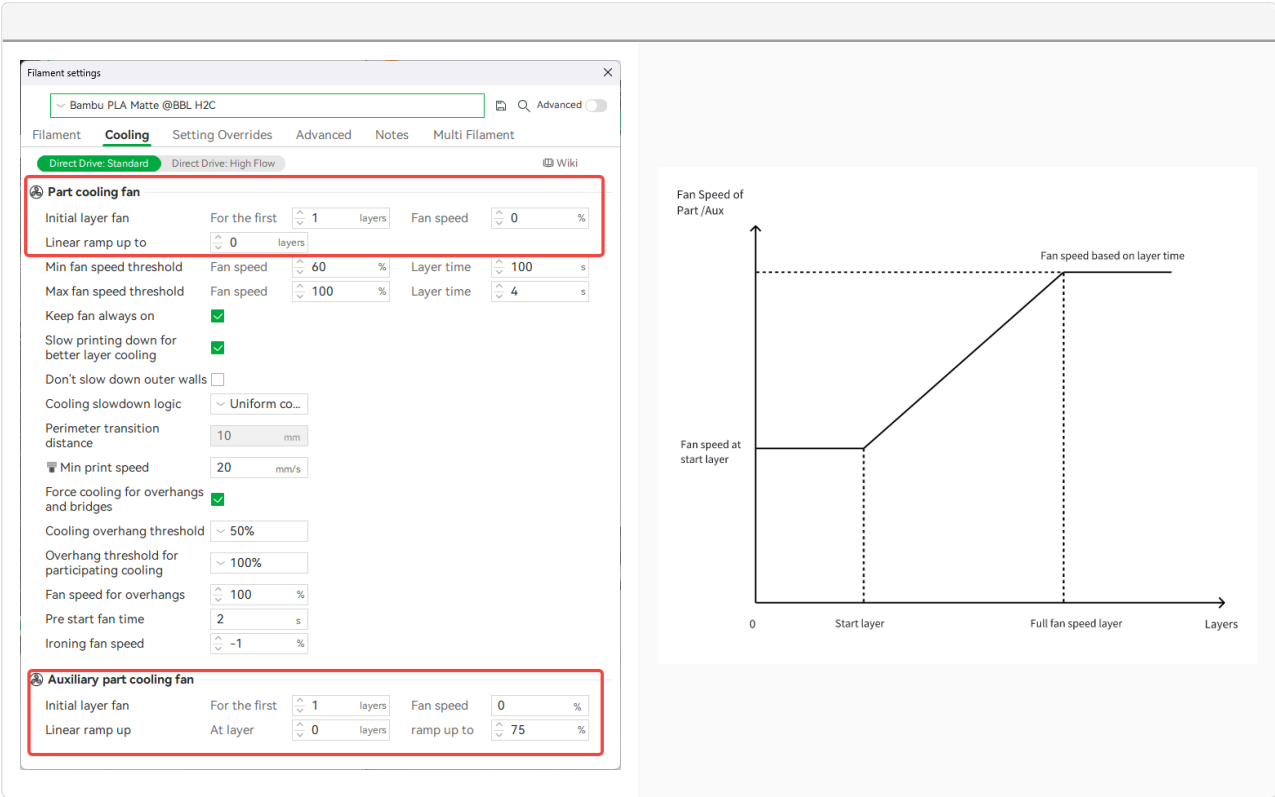
[Video tutorial](#)

[Learn more](#)

Minimum supported firmware version: 01.02.00.00(H2C)、01.01.00.00(X2D)

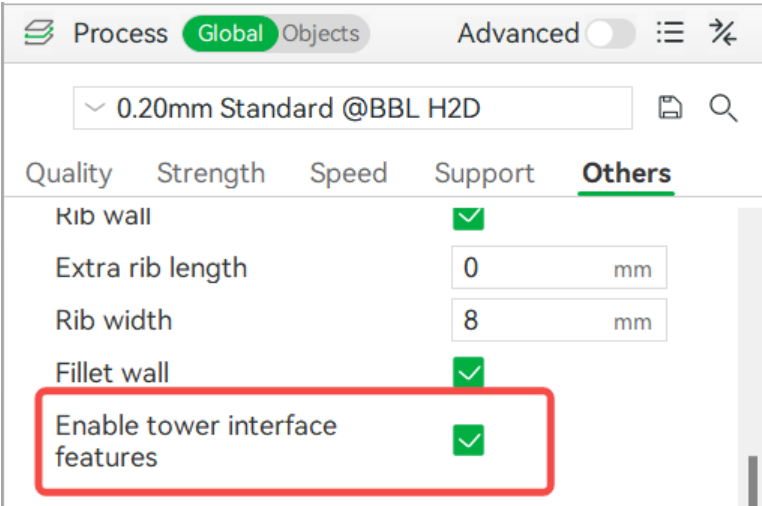
Linear Fan Speed Transition

Part cooling fan and auxiliary fan speeds now support more precise layer-by-layer control. Fan speed transitions linearly from the configured value at the starting layer to the target speed. This helps improve cooling conditions near the first layer, reducing the risk of warping or first-layer failures caused by sudden temperature drops.



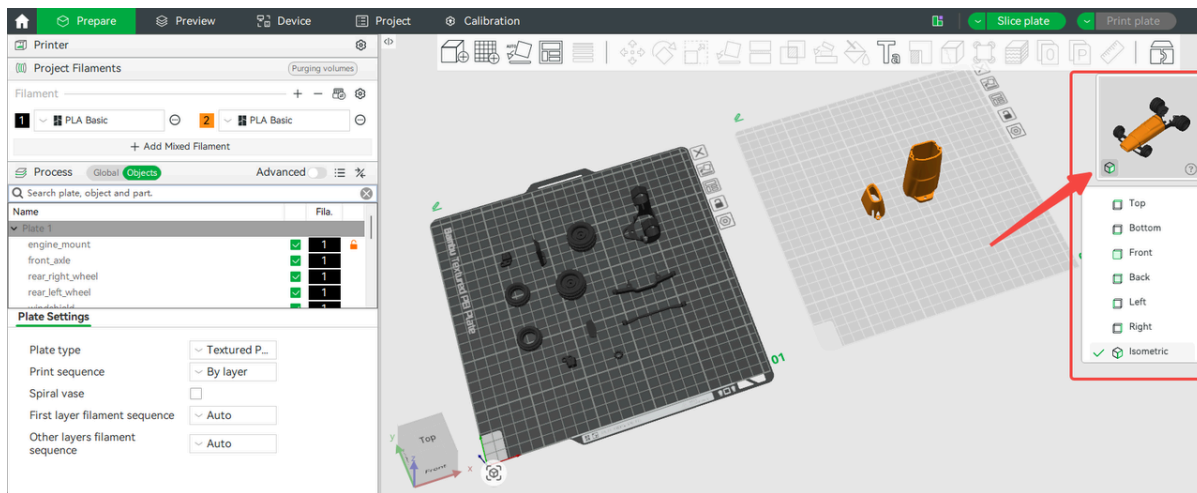
"Enable Tower interface features" available in Developer Mode for H2D/H2C/X2D

When enabled, it applies dedicated temperature, pre-extrusion, and flushing parameters to the prime tower contact layer, improving print quality during multi-material transitions and significantly reducing the risk of prime tower collapse when printing with filaments of significantly different viscosities.



Assembly Overview Preview

Added an Assembly Overview preview feature that allows users to preview the overall assembly effect in real time after changing colors or painting.



The preview angle can be adjusted using the viewpoint button in the View menu. This feature can be enabled or disabled in Preferences -> 3D Settings. View [Overview Tool](#) for more details.

3D Settings

- ☐ Zoom to mouse position
- ☒ Always show shells in preview
- ☒ Import a single SVG and split it
- ☐ Enable gamma correction for the imported obj file
- ☐ Remember last used color scheme

- Display overview Auto
- ☒ Improve rendering performance Auto
- ☒ Enable advanced graphics Open
- Toolbar Style Collapsed Close

Grabber scale

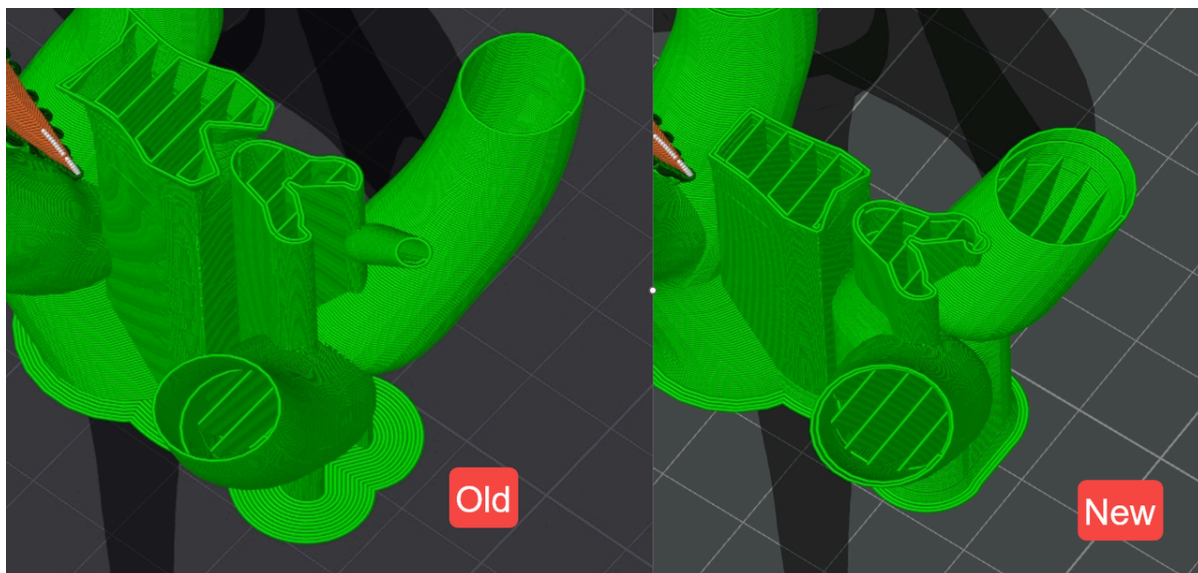
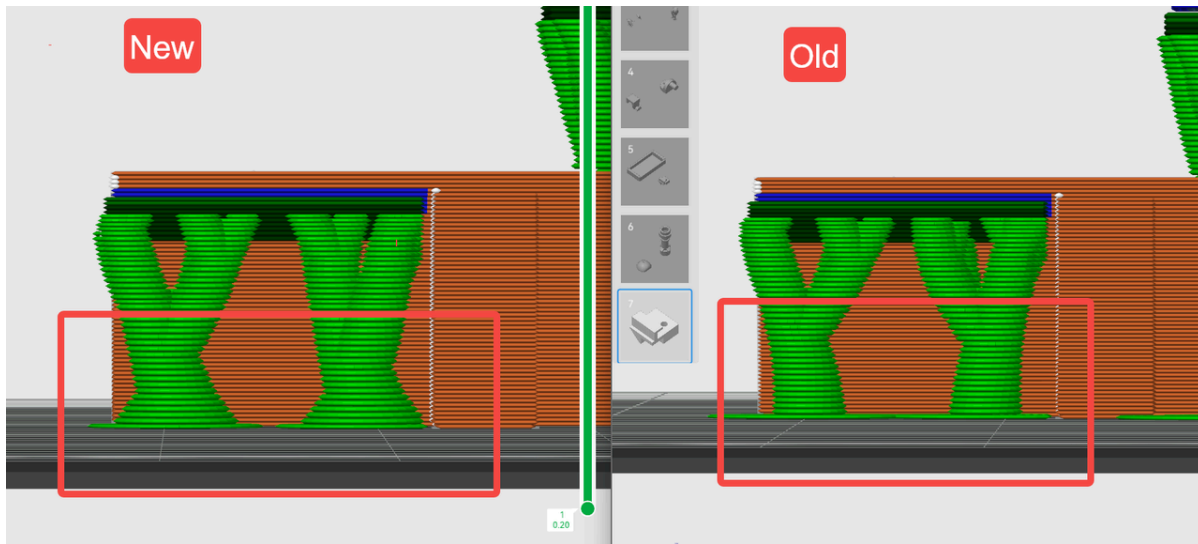
Tooltip offset

i Tip: For the best preview experience, it is recommended that 3MF creators import the complete assembled model file rather than individual separate parts.

Reduced Risk of Collapse During Printing

This update introduces optimizations from both support and prime tower perspectives to reduce the risk of collapse during printing:

- Wipe tower ironing is now enabled by default on P2S: In multi-material printing scenarios, this effectively improves adhesion between wipe tower layers, reducing the risk of wipe tower collapse.
- A chamfer has been added to the base of tree supports (expanding outward from a height of 2mm), effectively reducing the risk of support collapse. The radius calculation logic for support node merging has also been optimized, improving the consistency and accuracy of support structures.



Static Preview Frame on Liveview Connection or Switch

When Liveview is not auto-playing in Cloud or LAN mode, the Devices page now displays a static preview frame of the current device.

⚠ Minimum supported firmware version: 01.02.00.00 (P2S), 01.01.00.00 (X2D)



Fuzzy Skin: Additional Noise Types and Generator Modes

In addition to the Classic mode, Fuzzy Skin now supports 4 new noise types:

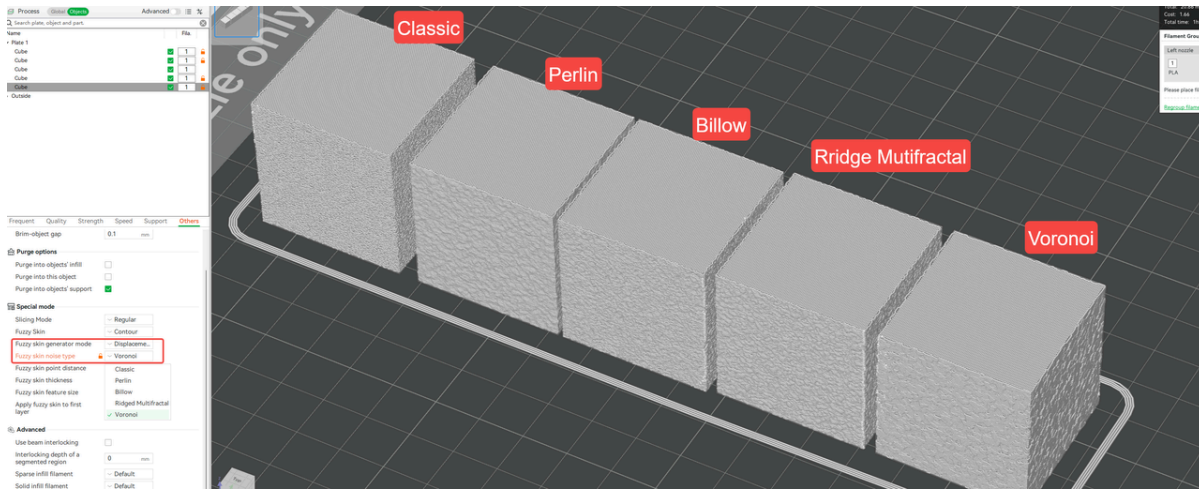
- Perlin
- Billow
- Ridged Multifractal
- Voronoi

New generator modes have also been added:

- Displacement: Creates texture by offsetting the nozzle left and right relative to the original path.
- Extrusion: Pattern is formed by varying the amount of extruded plastic (nozzle path stays straight).
- Combined: Displacement + Extrusion. Similar look to Displacement but fills gaps between perimeters.

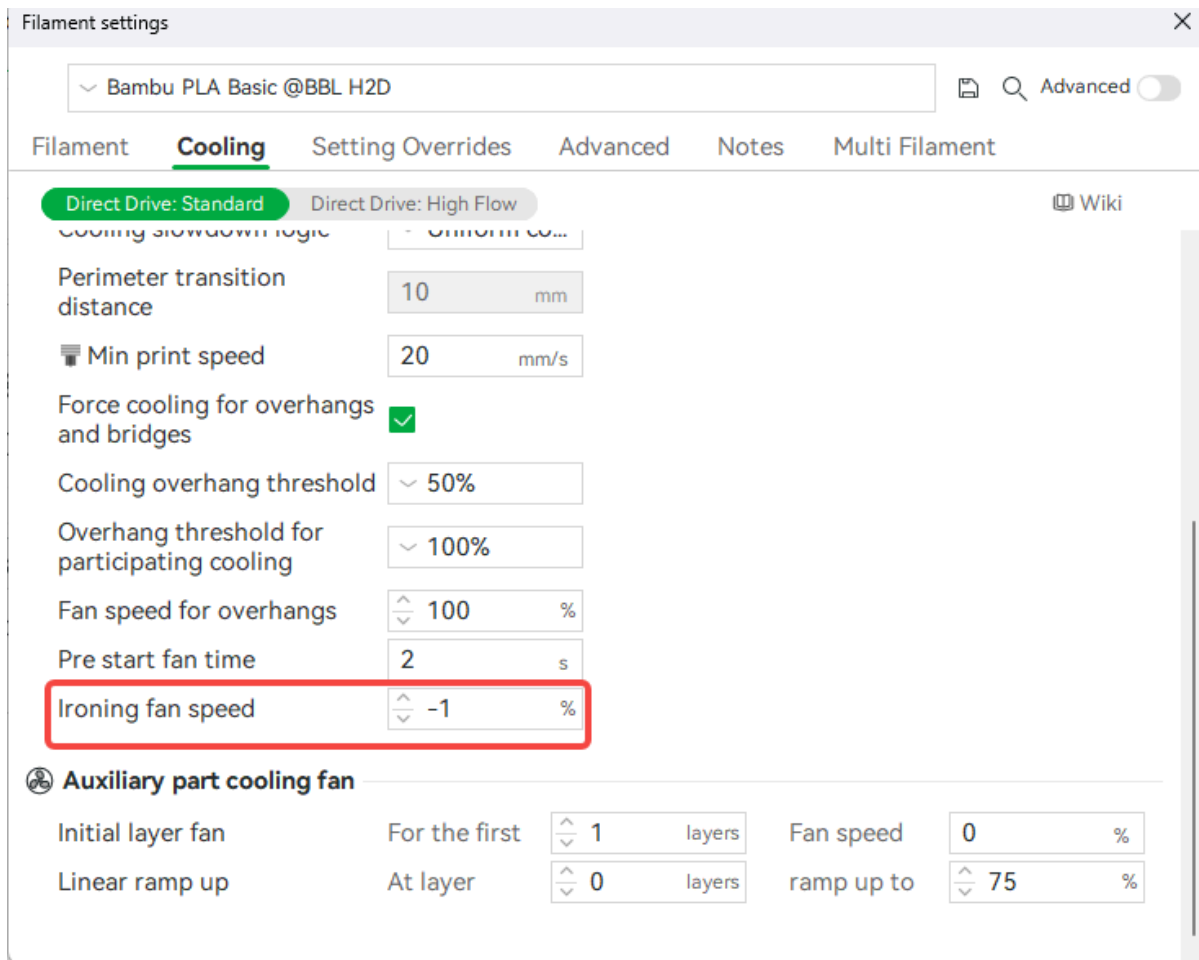
i Tip: Extrusion and Combined modes only take effect when using Arachne (variable line width) mode with "Fuzzy skin thickness" not exceeding actual line width.

Thanks to [@Arachnid](#) for the contribution.



Added Ironing Fan Speed Control

The ironing fan speed can now be configured independently. When set to -1, this feature is disabled and the fan speed follows the top surface fan speed; when set to any other value, the specified speed will be used exclusively during the ironing phase. This helps reduce material heat creep at low print speeds, resulting in a smoother ironing surface. Thanks to [@Noisyfox](#) for the contribution.



Helio Additive Update

This version upgrades the Helio integration with major performance improvements, clearer results and guidance, expanded compatibility handling, and improved reliability of workflows.

Product & Pricing updates (Alpha → Beta)

- ▶ Helio moves from Alpha to Beta with simplified pricing and practically unlimited Assess/Enhance under [fair use](#). The quota/token system has been deprecated.

- 2-week free trial with full access for new users.
- Product naming is now clearer:
 - Assess (formerly Simulation): predict where a print may fail before you print.
 - Enhance (formerly Optimization): autofix with new speeds; automatically optimizes for speed/strength/reliability/surface gloss.
- Improved activation flow.

Performance Improvements

Helio now runs with GPU acceleration for both Assess and Enhance, runtimes are reduced compared to previous versions (up to ~5× faster in supported scenarios).

Compatibility Handling & Safeguards

- Multi-material detection: Jobs containing multiple materials are now explicitly detected, with guidance provided to help users proceed safely.
- Unsupported material handling: When an unsupported material is detected, users are prompted to:
 - Select reference material for simulation, or
 - Switch to a supported material profile. Silent mismatches are no longer allowed.
- Improved matching for custom materials and printers:
 - Token-based fuzzy matching is used to identify the closest supported profiles.
 - Material-type fallbacks and confirmation prompts have been added.

Clearer and More Actionable Results

Assess results now more clearly explain why an issue is likely to occur, and which parameters or settings contribute to the risk.

- Fix suggestions are easier to locate and follow.
- The printability report has been refined to reduce ambiguity.
- Assessment guidance is more directly tied to concrete outcomes, such as improving reliability, reducing failure risk and increasing print speed.

Added Helio History

- View up to the last 10 Assess and Enhance runs.
- Revisit previous results and download generated G-code for comparison.
- Recent results can be recovered in case of network interruptions.

Reliability Improvements & Bug Fixes

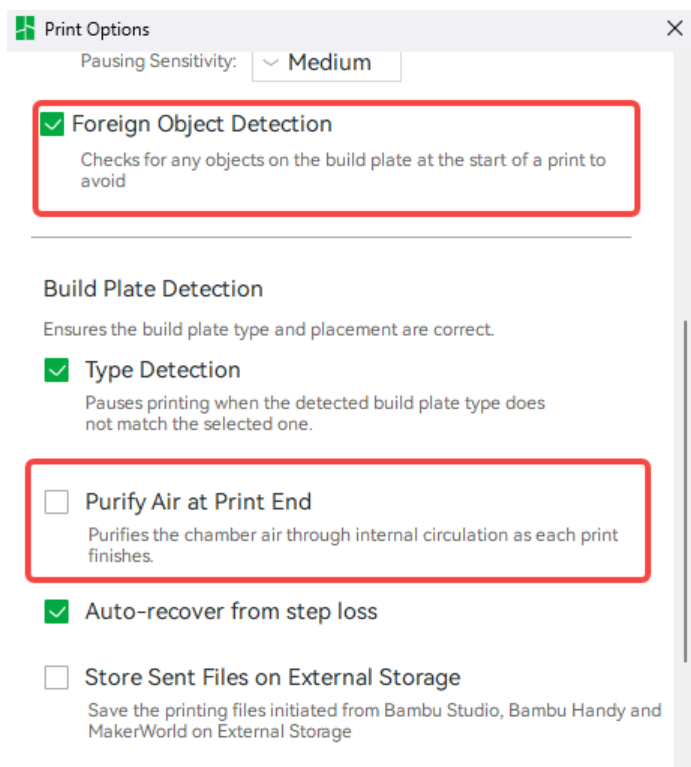
- Improved error detection during G-code polling.
- Backend error messages are now surfaced directly in failure dialogs, providing clearer diagnostics.

Availability & Limitations

- Helio is available on all Bambu Lab printers except X2D, based on current support status.
- For supported printers and materials, see <https://wiki.helioadditive.com/en/supportedprinters>  for more details.

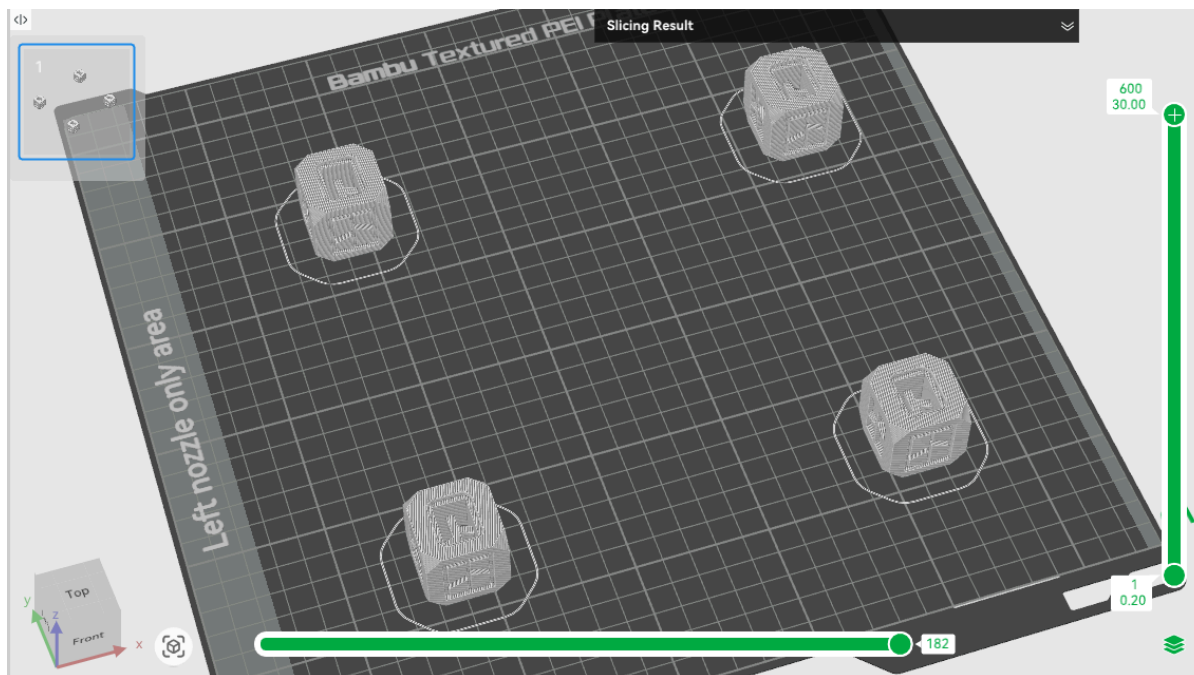
Improvements

Added support for Purify Air at Print End and Foreign Object Detection Options for P2S



Print-by-object optimization

- Interaction improvement: After enabling print-by-object, the interface automatically displays the print sequence with a prompt in the lower-right corner. The print sequence labels can be manually hidden.
- Each object now supports generating its own individual skirt. The skirt helps reduce first-layer extrusion artifacts after color changes, improving overall first-layer quality.
 - ⚠ Note: When this feature is enabled, it is recommended to leave an additional Skirt Distance of clearance between objects beyond the collision detection safety boundary, in order to reduce the risk of collision during printing.



Added cumulative print time display for H2C nozzle

The H2C nozzle info dialog allows users to monitor nozzle usage status.

Hotends Info

R		 Bambu PLA Basic 0.4 mm Standard Hardened	SN: 20D42A552600156 Version: 01.00.11.19	Used Time: 2 h
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H2C nozzle info dialog sync optimization

After syncing with an H2C printer, the nozzle information sync dialog will no longer reappear when switching pages, devices, or returning to the Prepare page after restarting.

Expanded “Reduce Infill Retraction” Modes

A new Auto option has been added. By default, this feature is enabled for most filaments to reduce print time; for stickier filaments, now for PETG, it is disabled by default to reduce the risk of filament debris sticking to the nozzle and causing print defects. The behavior of Disabled and Enabled remains unchanged, meaning the feature is always enabled or always disabled, respectively.

G-code output

Reduce infill retraction	v Auto
Add line number	Disabled
Filename format	<input checked="" type="button" value="Auto"/>
	Enabled

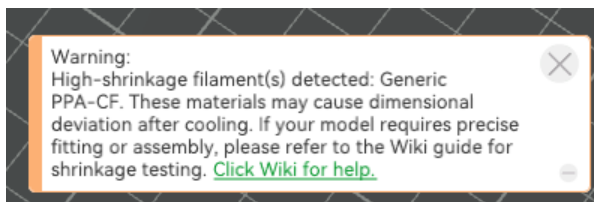
Post-processing scripts

Other Improvements

1. Device-side curve planning enhancement sync optimization: After syncing with a printer that supports curve planning enhancement, Bambu Studio will automatically disable arc fitting to ensure this feature functions correctly.

⚠ Minimum supported firmware version: 01.02.00.00 (H2C)

2. Added a slicing tip for high-shrinkage filaments.



3. H2S/H2C/P2S now support remote drying and dry-while-printing for AMS 2 PRO and AMS HT from Studio's Devices page.

⚠ Minimum supported firmware version: 01.01.40.00 (H2S), 01.02.00.00 (H2C), 01.02.00.00 (P2S)

4. H2C left extruder now supports TPU printing (TPU 90A/95A/95A HF).

⚠ Minimum supported firmware version: 01.02.00.00 (H2C)

5. Added nozzle clumping detection and calibration for H2S.

⚠ Minimum supported firmware version: 01.01.40.00 (H2S)

6. All printer models now support aborting the filament load/unload process.

⚠ Minimum supported firmware version: 01.01.40.00 (H2S), 01.02.00.00 (H2C), 01.01.50.40 (P2S), 01.09.01.00 (P1S), 01.07.01.00 (A1)

以下为beta补充内容，即github page不包含该部分，wiki需要包含

7. Improved the user experience of Change Type by turning it into a submenu with checkmarks for quicker and clearer selection. Thanks to [@jomixlaf](#) for the contribution.

8. Added model/step to the supported MIME type list on Unix systems, improving STEP model integration. Thanks to [@hadess](#) for the contribution.

9. Added a winget updater GitHub Action to improve packaging and distribution workflow. Thanks to [@LightDestory](#) for the contribution.

10. Updated the macOS CI environment from macos-13 to macos-15-intel for continued build reliability. Thanks to [@raulp](#) for the contribution.

11. Upgraded GitHub Actions. Thanks to [@salmanmkc](#) for the contribution.

Bug fixes

1. Fixed an issue where P2S liveview could drop frames in certain scenarios.
2. Fixed an issue where the UI displayed abnormally across multiple monitors in certain scenarios. ([#9627](#) )
3. Fixed an issue where speed parameters were not correctly updated to match the selected nozzle type after switching nozzle types and re-slicing on H2C. ([#9807](#) )
4. Fixed an issue where enabling Fuzzy Skin painting and Avoid Crossing Walls simultaneously caused incorrect travel moves in certain scenarios. ([#9837](#) )
5. Fixed an issue where the top surface flow ratio was incorrectly reset in certain scenarios. ([#9563](#) )
6. Improved the logic for migrating process presets with the same nozzle type across different printer models. ([#9105](#) )
7. Fixed an issue where enabling the prime tower outer-wall gap ("Process"- "Prime Tower"- "Skip points") could cause missing outer walls in some scenarios.
8. Fixed a conflict issue related to the user program under the web directory, thanks to @Bambu-Pomfret.
9. Beta update notification was enabled by default since this beta, which can be disabled in Preferences.
10. Fixed an issue where the top surface flow ratio in object settings could reset after re-selecting the object. ([#9563](#))
11. Fixed unreadable overlay elements in the 3D view on XWayland/Intel platforms. This fix comes from contributor @pedroslan.
12. Fixed an issue where the SwitchBoard control could not correctly register left-side mouse clicks when shown on a secondary monitor positioned to the left. Thanks to @mpaperno for the contribution.
13. Fixed a divide-by-zero exception. Thanks to @mpaperno for the contribution.
14. Fixed a CLI segmentation fault that could make command-line model workflows unusable. Thanks to @niccolodevries for the contribution.
15. Fixed a crash that could occur when slicing with Fuzzy Skin point distance set to 0 mm.
16. Fixed a memory leak caused by an allocated but uncleaned object instance. Thanks to @mpaperno for the contribution.
17. Added support for localized cuts. Thanks to @StateoftheeArt for the contribution.
18. Refactored GUI parameter input validation to support floating-point minimum and maximum values, instead of only integers. Thanks to @mpaperno for the contribution.
19. Fixed window drag lag on Windows. Thanks to @4RH1T3CT0R7 for the contribution.
20. Fixed an issue where acceleration settings on P2S could affect travel moves.

Known Issue

Due to a system change in macOS Tahoe 26, the default download path may be redirected to a system directory without write permission, which can cause MakerWorld file downloads to fail. If you encounter this issue, please go to Preferences and select a different save path to resolve it.

End Notes

We hope this guide has provided clear and practical support.

If the issue remains unresolved, please submit a [support ticket](#)  and include your recent printer logs and additional pictures or other details. Our technical team will review your request and provide detailed assistance.

You can also visit [Bambu AI](#) , which can instantly answer common questions and provide you with operational guidance.